

Multi-Model Consistency for LLMs' Evaluation

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Abstract—This paper introduces an evaluation method for large language models (LLMs) based on multi-model factual cognition consistency. Traditional evaluation methods, especially in terms of factuality assessments, face challenges in constructing extensive domain-specific question sets and relying on specific model answers. These methods fall short in the face of dynamic and diverse model development. To overcome these limitations, the proposed approach does not depend on a fixed set of standard answers. Instead, it utilizes the responses of multiple models to construct a dynamic, relative evaluation benchmark. We first developed a framework to capture and compare the cognitive consistency of different models when addressing specific questions. Subsequently, a dynamic iterative algorithm was designed to evaluate models based on these sets of answers. Experiments across multiple domains demonstrated the effectiveness of this method. This innovative evaluation strategy not only provides a more comprehensive and flexible approach to understanding and assessing the performance of LLMs in various scenarios but also offers practical guidance for future model development and improvement.

Index Terms—Large Language Models, Model Evaluation

I. INTRODUCTION

In the current landscape of artificial intelligence, large language models (LLMs) have garnered immense attention due to their powerful processing capabilities and broad application prospects [1]. These models have demonstrated exceptional performance across various tasks, sparking widespread interest in both academic and industrial circles [2]. However, as these models continue to evolve and find application, their assessment has become increasingly crucial. Evaluation not only aids in understanding the capabilities and limitations of models but also serves as a guide for further optimization and application. Particularly in assessing the factualness of LLMs, such evaluations are not only about whether models can generate statements consistent with the objective world but also a key dimension in measuring their practicality and performance [3].

Presently, assessments of factuality often involve posing a multitude of factual questions across different domains or using the responses of certain high-performing models as benchmarks for evaluating the abilities of others [4]. This approach can effectively assess the factual capabilities

of models to some extent. However, with the emergence of more models designed for specific scenarios, traditional evaluation methods face new challenges [5]. Firstly, methods reliant on specific question sets struggle to comprehensively assess model performance in particular domains due to the time-consuming and labor-intensive nature of constructing a large number of high-quality, domain-specific question-answer samples. Secondly, although some models excel in overall capabilities, this does not imply their ability to provide high-quality answers in all specific domains. Thus, relying solely on these models' answers as evaluation standards may lead to the failure of domain-specific assessments.

The core challenge lies in the fact that traditional evaluation methods depend on constructing static, fixed benchmarks, which might be ill-suited to accommodate the diverse development and application needs of models. Therefore, researching a dynamic, non-static model performance evaluation strategy is vital for adapting to various scenario-specific model assessments.

In light of this, this paper proposes an evaluation method based on multi-model factual cognition consistency (as shown in Figure 1). This method does not rely on a fixed set of standard answers but instead constructs a dynamic evaluation benchmark through the responses of multiple models. By comparing the consistency between the responses of the model under evaluation and those of multiple models, a relative, dynamic evaluation metric is established. The advantage of this method is its ability to showcase the relative capabilities of different models in specific domain performance evaluations and its broad adaptability.

Specifically, we first propose a framework capable of capturing and comparing the cognitive consistency of different models when addressing specific questions, obtaining a set of answers that facilitate easy comparison of consistency. Subsequently, we designed a dynamic, iterative algorithm to evaluate the cognitive consistency of multiple models based on these answer sets. Finally, experiments in multiple domains validated the effectiveness of this method. This innovative evaluation approach allows for a more comprehensive and flexible understanding and assessment of the real-world performance of LLMs in various application scenarios, providing practical guidance for their future development and improvement.

The contributions of this paper are summarized as follows:

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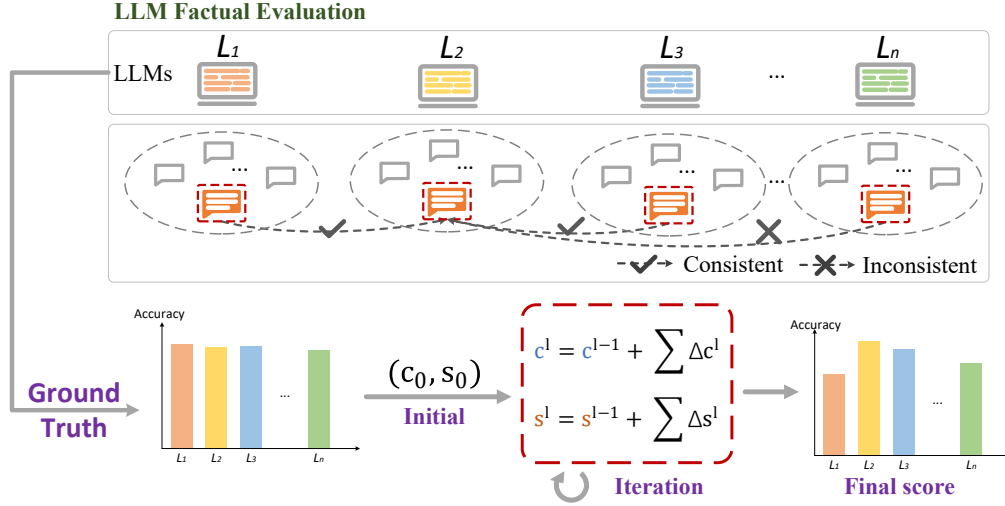


Fig. 1: LLM factual evaluation with consistency mechanism

- We introduce a dynamic evaluation benchmark for LLMs, employing multi-model factual cognition consistency, moving beyond traditional static benchmarks and enabling more flexible, scenario-specific assessments of model capabilities.
- We develop an innovative, iterative algorithm to assess cognitive consistency, allowing for continuous refinement and more accurate evaluations of LLMs, especially in correcting initial inaccuracies due to limited data.
- We demonstrate the robustness and effectiveness of the proposed method under various conditions, including different numbers of prior samples and reduced numbers of participating models, highlighting its adaptability and practical applicability in diverse scenarios.

II. RELATED WORKS

With the rapid development of LLMs such as ChatGPT and LLaMa-2 [6], how to quantitatively assess these large models has become a hot research topic in the current field of natural language processing. Firstly, evaluating LLMs contributes to a better understanding of their strengths and weaknesses. For example, benchmark tests like PromptBench [7] have demonstrated that current LLMs are highly sensitive to challenging prompts, highlighting the need for more refined prompts to achieve better performance. Secondly, improved evaluation methods can provide better guidance for human interaction with LLMs, inspiring future interaction design and implementation. Existing literature on the evaluation of LLMs typically categorizes evaluation methods into two categories: automatic evaluation and human evaluation.

Automatic evaluation is a common assessment method, typically involving the use of standard metrics and evaluation tools to assess model performance. Researchers such as Qin et al. [8] and Bang et al. [9] have utilized automated methods to assess a variety of tasks. With the continuous advancement of LLMs, more sophisticated automatic evaluation methods

have emerged. Lin and Chen [10] introduced LLM-EVAL, an innovative multi-dimensional automatic evaluation technique for open-domain conversations. In parallel, PandaLM [11] has been developed to enable replicable and automatic evaluations, where LLMs serve as evaluators or "judges" of other models. Additionally, Jain et al. [12] proposed a self-supervised evaluation framework that enhances efficiency by reducing the need for extensive data labeling.

Human evaluation serves as a supplementary approach to automatic evaluation. Based on relevant literature, six criteria for human evaluation can be identified: Accuracy [3], [13] primarily assesses the correctness of generated text, ensuring that the information provided by the language model aligns with factual knowledge. Relevance [14] focuses on the appropriateness and significance of generated content, ensuring that the text offers relevant and directly applicable information for specific contexts or queries. Fluency [15] evaluates whether the text generated by the model is smooth, maintains a consistent tone, and preserves stylistic consistency, avoiding unnatural expressions and abrupt topic transitions. Transparency concerns the clarity and openness of the model's decision-making process, including how the model conveys its thought process, enabling users to understand how its responses are generated and why. Safety [16], as a key criterion, involves assessing potential harm in generated content to ensure user well-being and information accuracy. Human consistency assesses the degree to which model outputs align with human values, preferences, and expectations, ensuring compliance with societal norms and user expectations.

Assessment methods for LLMs have shortcomings. Automatic evaluations, while objective and efficient, often overlook the complexities of natural language [17], [18], and are limited by the reliance on auxiliary models which may introduce faults [4]. Reference-based metrics like ROGUE [19] and BLEU [20] show low correlation with human judgments in complex tasks, as they focus on lexical overlap and miss semantic nuances

[21] [22]. Human evaluations, essential in specific domains like healthcare [23], suffer from inconsistency and bias [24], [25] [15]. They are also time-consuming and costly, needing many evaluations for significant conclusions [26], [27]. Krishna et al. highlight challenges in human evaluations, such as subjectivity and workload [28]. Future research should focus on developing innovative, objective, and consistent evaluation methods for open-ended tasks.

III. METHODOLOGY

In this section, we detail the process of evaluating LLM performance through logical question-and-answer interactions. First, we describe how to obtain processable responses from LLMs by constructing appropriate prompts. Then, we present an algorithm designed to evaluate the performance of various models based on their responses, along with some analysis of the algorithm.

A. Construction of Prompts for LLM Logical Testing

Logical inference based on learned knowledge and providing responses is a crucial ability of large models. Due to cost constraints, obtaining a large-scale annotated dataset for LLM testing and evaluation is often challenging. Many tests opt to compare with the best-performing large models (often OpenAI's GPT-4) as a benchmark. However, GPT-4 itself still has issues such as hallucinations, making this comparative evaluation not always reliable. Therefore, we have designed a large model evaluation scheme based on multi-source consistency. Given a set of variables under a pre-designed specific scenario D_X and their corresponding explanatory texts $X = \{x_1 : t_1, x_2 : t_2, x_3 : t_3, \dots, x_n : t_n\}$, we use the following Prompt to test the logical inference results of the LLM L_i :

*You are an expert in D_X , investigating the causal relationships between the following variables in your field. Please provide **all** the variables that affect x_i **directly** in the format of $\langle \text{Var} \rangle$ potential cause $\langle / \text{Var} \rangle$. Variables and their meanings are presented as follows. $\{x_1 : t_1; x_2 : t_2; \dots x_n : t_n\}$*

Key words are highlighted in bold, which make LLMs output as comprehensive causes as possible to cover the true causes. By processing the query response to this Prompt, the causal set of the variable x_j can be obtained:

$$r_{ij} = \mathcal{F}(L_i, Q(D_X, X; x_j)) \quad (1)$$

Here, $\mathcal{F}(\cdot)$ represents the method for extracting the variables provided by the LLM from the query response. r_{ij} denotes the causal set of x_j given by L_i . By using the Prompt Q to query the model multiple times, the causal set r corresponding to each variable in the dataset is determined, and r is used as the result of the model test for its evaluation.

B. Evaluation of Model Logical Ability Based on Consistency

Without loss of generality, assume that a set of LLMs to be tested is denoted as $L = \{L_i | i \in 1, 2, \dots, N\}$ and a set of questions is denoted as $Q = \{Q_j | j \in 1, 2, \dots, M\}$. Let r_{ij} denote the response of LLM L_i to question Q_j . All other response to Q_j made by other LLMs in L except L_i are collected into a set R_j . This can be formally expressed as:

$$R_{ij} = \{r_{kj} | k \neq i\} \quad (2)$$

This paper proposes a method to comprehensively measure the performance of large models using two metrics, including consistency C and stability S . For a LLM L_i , it can be represented as:

$$\mathcal{P}(L_i) = (C_i, S_i) \quad (3)$$

$$\mathcal{P}(r_{ij}) = (c_{ij}, s_{ij}) \quad (4)$$

This involves $C_i = \frac{1}{M} \sum_{j=1}^M c_{ij}$ and $S_i = \frac{1}{M} \sum_{j=1}^M s_{ij}$, where c_{ij} and s_{ij} represent the consistency and stability scores, respectively, for the response r_{ij} given by L_i to the question Q_j .

In the process of comparative evaluation, consistency represents an assessment of whether the overall quality of a model's answers aligns with reality, while stability characterizes the fluctuation in the quality of the model's answers to different questions. It should be noted that the stability here characterizes the fluctuation range of the model's answer quality. Therefore, a larger value indicates greater fluctuation in the quality of the model's answers, meaning poorer stability in the traditional sense. During the testing process, based on the score of each model itself, the quality of answers from different models to each question is evaluated, and the model scores are iteratively updated based on the evaluation results.

This method of evaluating a model's logical performance by using the consistency statistics of different models' answers to the same question can be considered as a process of statistical observation. LLMs inherently exhibit a certain degree of randomness in generating responses, and this randomness affects the quality of the model's answers. In many application scenarios, such as finance, healthcare, and military, we hope that LLMs can provide stable and reliable answers, thus necessitating the measurement of the model's randomness to control risks.

A method founded on the fusion of consistency comparison and prior evaluation is proposed to iteratively update the score of LLMs' logical capability. The iterative update of the LS of L_i is represented as follows:

$$\mathcal{P}^l(r_{ij}) = \mathcal{E}(\mathcal{P}^{l-1}(r_{ij}); \{\mathcal{P}^{l-1}(r_{kj}) | r_{kj} \in R_{ij}\}; A_i) \quad (5)$$

Here, $\mathcal{E}(\cdot)$ signifies the model score update method, and l denotes the l^{th} iteration. $\mathcal{P}^l(r_{ij})$ represents the score of model L_i in the l^{th} iteration for question Q_j . To initialize ($l = 0$), some questions Q_{label} with standard answers G are provided as ground truth to LLMs. The accuracy of LLMs in responding to these questions will serve as a prior model score.

$$\begin{cases} C_0 = \mathcal{F}(L_i, Q_{label}, G) & i \in 1, 2, \dots, N \\ S_0 = 0 \end{cases} \quad (6)$$

$$\mathcal{P}^0(r_{ij}) = (C_0, S_0) \quad (7)$$

The approval of the response r_{ij} of the model L_i to domain-specific question Q_j compared to the responses R_{ij} of other models is denoted by A_i . It is defined as a set $A_i = \{a_k | r_{kj} \in R_{ij}\}$, where a_i is determined by:

$$a_i = \begin{cases} 1 & r_{kj} = r_{ij} \\ 0 & r_{kj} \neq r_{ij} \end{cases} \quad r_{kj} \in R_{ij} \quad (8)$$

In this expression, $r_{kj} = r_{ij}$ implies that models L_k and L_i provide consistent answers to question Q_j . In other words, the sets of relationships they provide for the variable x_j are the same. During each iteration, the consistency and stability of r_{ij} is updated based on the consistency comparison, with the initial value serving as the prior. The outcome is then used as the prior in the next iteration.

During the l^{th} iteration of (5), the update for the consistency c_{ij} of model L_i is designated as follows:

$$c_{ij}^l = c_{ij}^{l-1} + \sum_{r_{kj} \in R_{ij}} \Delta c_{i,k}^l \quad (9)$$

$$\Delta c_{i,k}^l = \frac{\mu_{i,k}^{l-1} \times \beta_{i,k}^{l-1}}{N} \quad (10)$$

Here, $\Delta c_{i,k}$ represents the update amount, where $\mu_{i,k}^{l-1}$ signifies the update range, the positivity or negativity of which is determined by the consistency comparison of the results. $\beta_{i,k}^{l-1}$ denotes the update magnitude, a positive value derived from the stability metric of the models.

The precise description of $\mu_{i,k}^{l-1}$ in (10) is as follows:

$$\mu_{i,k}^{l-1} = a_i - c_{ij}^{l-1} \quad (11)$$

Through the definition of μ , when a model's response is consistent with others, i.e., $r_{ij} = r_{kj}$, its consistency score for that question will increase (noticed that $c_{ij}^{l-1} < 1$), and vice versa. The magnitude of change is determined by the combined prior of the consistency and stability scores of the two models' responses, which is reflected in the design of $\beta_{i,k}^{l-1}$:

$$\beta_{i,k}^{l-1} = \frac{1}{2} (1 + (c_{kj}^{l-1} + c_{ij}^{l-1}) - (s_{kj}^{l-1} + s_{ij}^{l-1})) (c_{kj}^{l-1} + s_{kj}^{l-1}) \quad (12)$$

For an LLM L_i , as mentioned above, c_{ij} represents the answer quality measured by consistency, and the difference between it and stability s_{ij} reflects the range of variation in this quality, similar to variance in statistics.

It can be observed that, during the iterative process, the consistency score of a model is jointly influenced by its prior score and the scores of other models. Such a design ensures that models with higher consistency scores undergo smaller updates in their scores, making them less susceptible to the influence of other models' response quality. On the other hand,

these better-performing models also possess greater weight in influencing other models.

For the calculation of stability s , a similar iterative process can be executed:

$$s_{ij}^l = s_{ij}^{l-1} + \sum_{r_{kj} \in R_{ij}} \Delta s_{i,k}^l \quad (13)$$

$$\Delta s_{i,k}^l = \frac{\alpha_{i,k}^{l-1} \times \zeta_{i,k}^{l-1}}{N} \quad (14)$$

where $\alpha_{i,k}^{l-1}$ represents the update amount and $\zeta_{i,k}^{l-1}$ illustrates the update range in the l^{th} . The design of $\alpha_{i,k}^{l-1}$ is expressed as:

$$\alpha_{i,k}^{l-1} = N \frac{s_{ij}^{l-1} \times \Delta c_{i,k}^l}{c_{ij}^{l-1}} \quad (15)$$

It should be noted here that the sign of α is the same as that of μ , determined by whether r_{ij} and r_{kj} are consistent. The equation is derived from dividing the amount by which c_{ij} is updated in each iteration by its value, i.e., the update rate of c_{ij} , ensuring that the update speed of s_{ij} does not exceed that of c_{ij} . Their combined effect ensures that stability and consistency exhibit similar but more moderate tendencies to change. The design of $\zeta_{i,k}^{l-1}$ is expressed as:

$$\zeta_{i,k}^{l-1} = \frac{2}{\pi} \arctan(\lambda |c_{kj}^{l-1} - c_{ij}^{l-1}| - 1) \quad (16)$$

$\arctan(\cdot)$ is utilized for regularization. Here, λ is a hyper-parameter, and ζ is a positive or negative number depending on the difference in consistency between the two models derived from the previous iteration process. Due to the characteristics of the $\arctan(\cdot)$ function, the hyper-parameter λ controls the threshold for updating stability. When λ is large, it becomes easier to perceive a discrepancy in the consistency scores between two models, leading to a greater impact on the final estimate of consistency c . Conversely, with a smaller λ , it becomes more challenging to influence the change in c .

The underlying mechanism of the model stability assessment constructed from the above equation is as follows. When two models provide the same answer, α is positive. In this situation, if there is a discrepancy in the consistency scores of the two models, ζ is positively correlated with this discrepancy, leading to an increase in s_{ij} as the discrepancy grows. If all models provide consistent answers, it indicates that they have the same performance on that question, and they will obtain the same consistency score through iteration.

IV. EXPERIMENTS

A. Experimental setup

1) *Dataset*: The dataset utilized in the experiment is derived from a task that focuses on identifying semantic causal relationships between given concepts and their corresponding descriptive texts. The concepts are extracted from a causal network, intricately designed by experts in the field¹. Variables,

¹Resources could be obtained from <https://www.bnlearn.com/bnrepository/>.

TABLE I: Description of utilized LLMs.

LLM	# Parameter	Developer	Year
GPT-3.5	175B	OpenAI	2022
PaLM	540+B	Google	2022
GPT-4	220B*8	OpenAI	2023
Claude-2	130+B	Anthropic	2023
LLaMa-2	70B	Meta	2023

each accompanied by their detailed meanings, contextual background information, and crafted prompts, are amalgamated to construct a robust set of causal questions. The ground truth answer is determined by the presence or absence of a connection between the two variables in the causal network. Each question-answer pair undergoes a rigorous manual review process to ensure semantic clarity and accuracy. Pairs that fail to meet the criteria of being easily comprehensible and appropriate for human understanding are excluded. The final dataset comprises 212 question-answer pairs.

In order to streamline the extraction of definitive results, a single-choice prompt has been developed. This design aims to elicit a distinct response from LLMs regarding the presence or absence of a causal relationship.

2) *LLMs to be evaluated*: Five popular LLMs are included in the experiment for evaluation. 1) GPT-3.5 [29]: Developed by OpenAI, GPT-3.5 is an upgraded version in the GPT-3 series. It enhances text generation accuracy and fluency over its predecessor, GPT-3. 2) PaLM [30]: A LLM by Google’s research team, PaLM, short for “Pathways Language Model,” is built on Google’s Pathways system and aims to handle complex language understanding and generation tasks. 3) GPT-4: Another LLM from OpenAI, GPT-4 is a successor to GPT-3. It features further optimization and expansion in model architecture, training data, and algorithms, enhancing performance across various language tasks. 4) Claude-2 [31]: Developed by Anthropic, a company formed by former OpenAI employees, Claude-2 is one of their models designed to tackle specific language processing challenges. 5) LLaMa-2 [6]: The latest open-source large model released by Meta AI. LLaMa-2’s hallmark is a doubled training data size of 2 trillion tokens and a twofold increase in context length limits, crucial for large model usage. Detailed information about these LLMs is presented in Table I.

3) *Parameter settings*: The evaluation result is reported after the proposed evaluation method completes its 50 iterations. To estimate the initial accuracy of the LLMs at the beginning of the proposed method, a small set of samples is utilized as a prior. The default number of prior samples used for this estimation is 20.

The experiments are carried out on a server equipped with a 12th Gen Intel Core i5-12400 CPU and 16GB of RAM, operating under Ubuntu 20.04.5 LTS.

4) *Evaluation metrics*: Accuracy is the simplest and most direct evaluation metric. It can be defined as the ratio of correctly predicted instances to the total number of instances. In other words, it measures the proportion of true results (both true positives and true negatives) among the total number of

cases examined. This metric is particularly valuable in scenarios where the outcomes are equally important, providing a clear and concise measure of the model’s overall effectiveness in predicting the outcomes correctly.

The accuracy obtained from the proposed evaluation model is not the actual accuracy, but an estimated accuracy derived through the method. More intriguing than the absolute values of accuracy is the accuracy of the proposed method in ranking several models. This approach more directly characterizes the merits and demerits of the models and eliminates the interference from the intrinsic difficulty of the tasks themselves. Consequently, a ranking-based metric is probably more suitable.

Let each LLM be indexed by i , and sorted them based on the estimated accuracy. The relationship between these rankings and a reference ranking (e.g., based on ground truth or another evaluation criterion) is quantified using the Spearman correlation coefficient. The coefficient, denoted as ρ , is calculated as follows:

$$\rho = 1 - \frac{6 \sum_{i=1}^n d_i^2}{n(n^2 - 1)} \quad (17)$$

where d_i is the difference in ranks of the i^{th} model between the two ranking systems, and n is the total number of models.

Furthermore, to assess the statistical significance of the observed correlation, we compute the p-value associated with ρ . A lower p-value (commonly, $p < 0.05$) indicates that the correlation is not merely a product of random chance.

B. Results and analysis

This section conducts experiments on the dataset, which answers the following research questions accordingly.

RQ1: How does the effectiveness of the proposed method compare to the estimated accuracy obtained using priors and the accuracy calculated using all samples?

RQ2: Does each iteration of the proposed method lead to a convergence towards a more accurate overall evaluation result?

RQ3: How does the proposed method perform under different conditions of prior numbers?

RQ4: Does the proposed method remain effective when the number of models is reduced?

1) *Result comparison (RQ1)*: To thoroughly investigate the effectiveness of the proposed evaluation method, three key areas are mainly focused on: a) How far the assessment has progressed from the initial accuracy estimated from priors. b) The remaining gap between our method and the precise outcomes derived from using all samples. c) The extent of the effectiveness and advantage of our proposed method compared to merely using prior estimations.

The study documents the accuracy estimates from the priors (denoted as ‘Prior’), the final accuracy predicted by our proposed method (denoted as ‘Final’), and the actual accuracy obtained from precise testing with all samples (denoted as ‘All’). A Spearman’s rank correlation test is then conducted between the rankings of these results and the rankings derived from ‘All’ data. The findings are presented in Table II.

TABLE II: The final evaluation compared with prior and all.

LLM evaluation	Prior		Final		All	
	accuracy	rank	estimate acc	rank	accuracy	rank
GPT-3.5	95.00	1	93.90 $\pm$ 2.00	2	89.15	2
PaLM	85.00	3	89.70 $\pm$ 1.93	4	83.49	4
GPT-4	90.00	2	94.67 $\pm$ 1.95	1	91.04	1
Claude-2	70.00	5	83.79 $\pm$ 1.75	5	80.66	5
LLaMa-2	80.00	4	91.98 $\pm$ 1.87	3	87.26	3
Spearman's rho with 'all'						
	correlation	p	correlation	p	correlation	p
	0.80	0.10	1.00	0.00	1.00	0.00

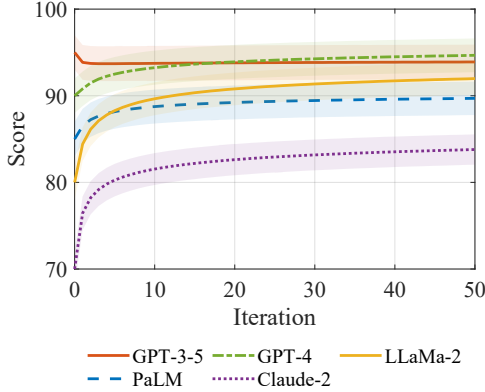


Fig. 2: LLM score converging with iteration

The experimental results address the three focal points outlined above. Firstly, the proposed method demonstrates a divergence from the initial prior accuracy and additionally provides a standard error to indicate the degree of uncertainty. The Spearman’s rank correlation test particularly sheds light on the comparative efficacy of the proposed evaluation method. Notably, the perfect correlation ($\rho = 1.00$) with a p-value of 0.00 for the ‘Final’ evaluation indicates a strong alignment with the ‘All’ data rankings, underscoring the robustness of the proposed method. The advancement from ‘Prior’ to ‘Final’ demonstrates significant improvement in ranking accuracy, proving the method’s capability to refine initial estimates effectively.

2) *Influence by number of iteration (RQ2)*: To verify the smoothness of the model convergence and the correctness of the convergence direction, as well as to explore the impact of the number of iterations in the proposed method on its actual effectiveness, we documented the scores of each LLM at different iteration counts during a single experiment, as shown in Figure 2.

As the iterations progress, it can be observed that the scores of the five LLMs gradually converge toward a specific final score in a manner that is initially rapid and then slows down. The convergence process is smooth.

Additionally, at iteration zero, the initial ranking of the LLMs’ performance is incorrect. Specifically, due to the limited number of samples used to estimate the prior accuracies, the accuracy of GPT-3.5 is overestimated, while the accuracy of LLaMa-2 is underestimated. During the iterations

of the proposed method, the estimated accuracy of LLaMa-2 rapidly increases, surpassing the PaLM model within ten iterations. Conversely, GPT-3.5, being overestimated initially, experiences a decline in its estimated accuracy as the iterations proceed and is overtaken by GPT-4 before the 20th iteration. This illustrates that the iterative process of the proposed model effectively corrects the errors in the prior estimates through consistent estimation.

3) *Influence by number of prior (RQ3)*: As mentioned above, due to insufficient sample size, prior estimates are prone to errors. To address this, different sample sizes were used to initialize the prior accuracy estimates. The evaluation results of the LLMs obtained by running our method with these estimates were then recorded. Additionally, the Spearman’s rank correlation test was conducted between the rankings of these results and those obtained from precise estimation using all samples. The test outcomes are presented in Table III, where cells with a grey background indicate better performance.

The experimental results demonstrate that the proposed method exhibits remarkable robustness. Specifically, when the number of priors reaches 15, the method is already capable of inferring the correct performance ranking of the LLMs, even though the priors still contain errors. Furthermore, even with as few as 5 priors, the method achieves a Spearman correlation coefficient of 0.90. This indicates that despite using a minimal number of initial samples and deriving highly inaccurate prior accuracy estimates, the proposed model still possesses strong capability to yield relatively accurate performance rankings of the LLMs.

4) *Influence by number of LLM to be evaluated (RQ4)*: Given that the proposed method is derived from mining and analyzing the consistency among various LLMs, an intriguing question arises: Would reducing the number of LLMs involved in the evaluation affect the results? To address this, we removed one LLM at a time from the original set of five used in the initial experiments, then ran the proposed method using only the remaining four LLMs. Subsequently, we plotted the curves showing the changes in the estimated LLM Scores as the method iterated. The results are depicted in Figure 3.

Recalling our previous precise results calculated using all samples, where the ranking was GPT-4 > GPT-3.5 > LLaMa-2 > PaLM > Claude-2 (corresponding to green > red > yellow > blue > purple in the graph), it is observed that even with the absence of one LLM, the proposed method, after 50 iterations, consistently achieves the correct performance ranking in all five scenarios. This reflects the method’s effectiveness even with fewer LLMs participating in the evaluation.

An more extreme scenario to consider is the effectiveness of the method with fewer LLMs involved, such as only three, or even more extremely, just two. The question then arises: does the proposed method remain effective in these more limited configurations? In Figure 4, only three LLMs are retained for model evaluation. Upon close examination, it is found that among the ten combinations, only the combinations of GPT-3.5 + PaLM + GPT-4 and PaLM + Claude-2 + LLaMa-2 exhibit minor errors in ranking order. In the more extreme

TABLE III: Evaluation with different prior numbers.

Prior Number	Method	GPT-3.5		PaLM		GPT-4		Claude-2		LLaMa-2		Spearman's rho	
		accuracy	rank	accuracy	rank	accuracy	rank	accuracy	rank	accuracy	rank	correlation	p-value
5	Prior	100.0	2	100.0	2	100.0	2	80.0	5	100.0	2	0.71	0.18
	Final	94.22 $\pm$ 2.04	3	92.55 $\pm$ 2.05	4	96.45 $\pm$ 2.03	1	85.59 $\pm$ 1.82	5	95.95 $\pm$ 2.04	2	0.90	0.04
10	Prior	90.0	1	90.0	1	80.0	3	60.0	5	80.0	3	0.32	0.60
	Final	92.55 $\pm$ 1.97	1	91.11 $\pm$ 1.99	4	92.31 $\pm$ 1.87	2	82.59 $\pm$ 1.68	5	91.95 $\pm$ 1.88	3	0.90	0.04
15	Prior	93.33	1	80.0	3	86.67	2	60.0	5	73.33	4	0.80	0.10
	Final	94.01 $\pm$ 1.99	2	88.95 $\pm$ 1.89	4	94.21 $\pm$ 1.92	1	83.22 $\pm$ 1.68	5	91.03 $\pm$ 1.81	3	1.00	0.00
20	Prior	95.0	1	85.0	3	90.0	2	70.0	5	80.0	4	0.80	0.10
	Final	94.11 $\pm$ 2.00	2	89.83 $\pm$ 1.93	4	94.96 $\pm$ 1.95	1	84.53 $\pm$ 1.75	5	92.31 $\pm$ 1.87	3	1.00	0.00
25	Prior	96.0	1	84.0	3	92.0	2	68.0	5	84.0	3	0.87	0.05
	Final	94.35 $\pm$ 2.01	2	89.92 $\pm$ 1.91	4	95.50 $\pm$ 1.96	1	84.97 $\pm$ 1.73	5	93.10 $\pm$ 1.90	3	1.00	0.00
30	Prior	96.67	1	86.67	3	93.33	2	73.33	5	86.67	3	0.87	0.05
	Final	94.24 $\pm$ 2.01	2	90.21 $\pm$ 1.94	4	95.70 $\pm$ 1.97	1	85.45 $\pm$ 1.78	5	93.52 $\pm$ 1.93	3	1.00	0.00
35	Prior	97.14	1	88.57	3	94.29	2	74.29	5	88.57	3	0.87	0.05
	Final	94.16 $\pm$ 2.01	2	90.41 $\pm$ 1.95	4	95.82 $\pm$ 1.98	1	85.83 $\pm$ 1.78	5	93.83 $\pm$ 1.94	3	1.00	0.00
40	Prior	92.5	1	85.0	3	90.0	2	72.5	5	85.0	3	0.87	0.05
	Final	92.97 $\pm$ 1.98	2	89.5 $\pm$ 1.93	4	94.78 $\pm$ 1.95	1	85.09 $\pm$ 1.78	5	92.95 $\pm$ 1.91	3	1.00	0.00
All		89.15	2	83.49	4	91.04	1	80.66	5	87.26	3	1.00	0.00

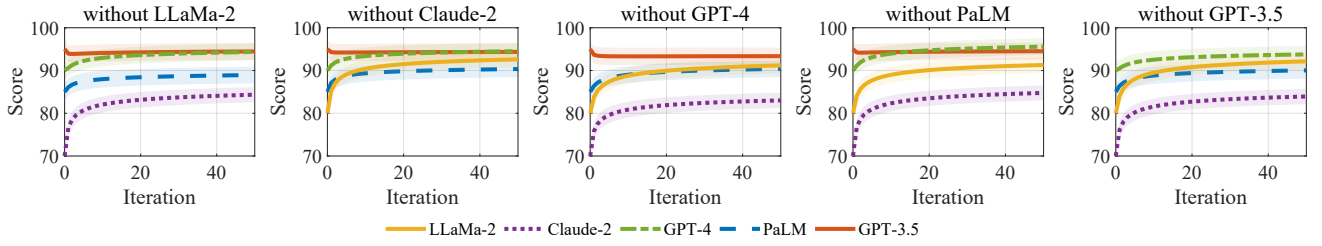


Fig. 3: Convergence with iteration when removing one LLM from the LLM list.

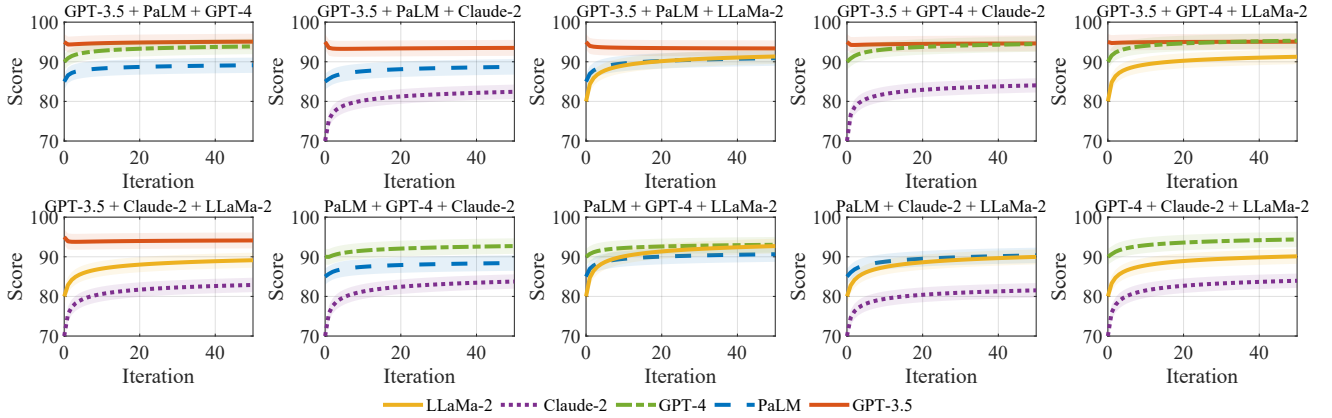


Fig. 4: Convergence with iteration when remaining only three LLM from the LLM list.

scenario depicted in Figure 5, where only two LLMs are retained, a thorough check reveals that among the ten combinations, errors occur only in the GPT-3.5 + GPT-4 and PaLM + LLaMa-2 pairings. This suggests that errors are primarily due to inaccuracies in the prior estimates, highlighting that the quality of priors may become even more crucial when fewer LLMs participate in the evaluation.

V. CONCLUSION

This paper explores an evaluation method for LLMs based on factual knowledge consistency, proposing a dynamic eval-

uation system that utilizes the relativity of multi-model answer consistency. Experiments were conducted in multi-source scenarios, analyzing the impact of data and the number of models on the effectiveness of the established evaluation system. The results demonstrate that the proposed method can effectively assess the factual logical cognition ability of LLMs in specific domains without ground truth. This explores a new way of evaluating LLMs, proving the feasibility of using multi-model answer consistency for dynamic quality assessment. Based on this study, how to use existing large-scale unlabelled knowledge data for low-cost, efficient evaluation of large

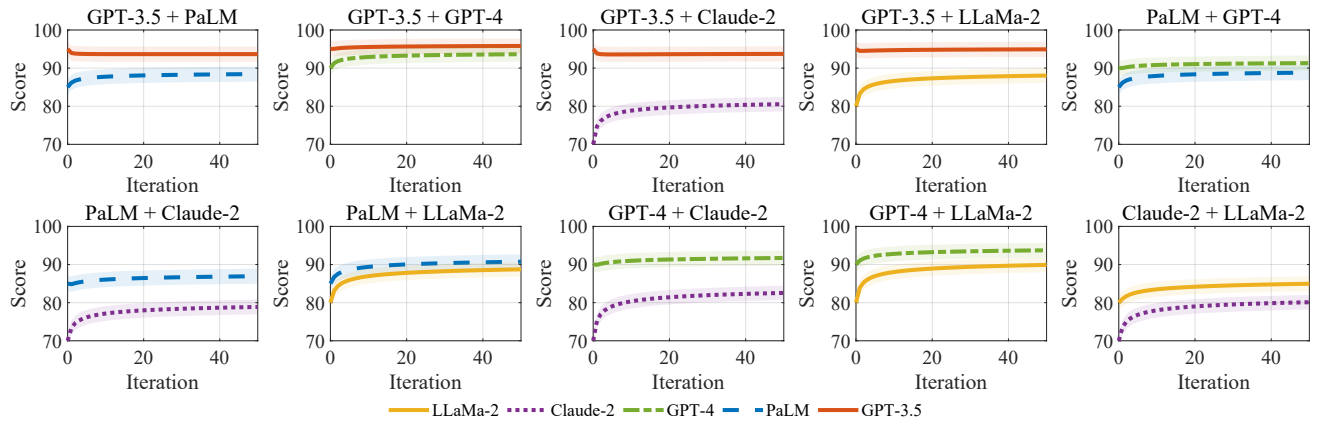


Fig. 5: Convergence with iteration when remaining only two LLM from the LLM list.

models may hold significant research value in the future.

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